

Language Modelling with Selective State Spaces

Project in Deep Machine Learning

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MSc course in Data Science and Artificial Intelligence

A.Y. 2025/2026



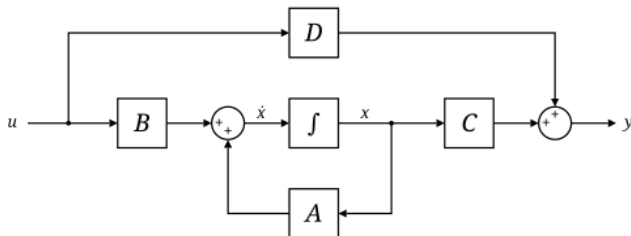
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State Space Models

With State Space Models (SSM) we mean hereinafter a type of linear model for sequences which keeps the current state of the “system” in a variable.

$$\dot{\mathbf{x}}(t) = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{B}(t)\mathbf{u}(t)$$

$$\mathbf{y}(t) = \mathbf{C}(t)\mathbf{x}(t) + \mathbf{D}(t)\mathbf{u}(t)$$



State Space Representation (from Wikipedia)

Discrete-Time SSMs

From the general form of the differential equation, we can derive the update formula for a discrete time system:

$$\begin{aligned}h_t &= \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t \\y_t &= \mathbf{C}h_t\end{aligned}$$

where $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$ come from a discretization step (zero-order hold):

$$\bar{\mathbf{A}} = \exp(\Delta\mathbf{A}) \quad \bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B} \approx \Delta\mathbf{B}.$$

Selective SSMs

Making $\mathbf{B}$, $\mathbf{C}$ and Δ input-dependent should lead to a stronger performance due to possible context filtering throughout the steps.

B , L , N and D identify the batch size, the length of the seq. the state-space dimension and the nr. of features, respectively.

Algorithm SSM + Selection (S6)

Require: $x : (B, L, D)$

Ensure: $y : (B, L, D)$

1: $\mathbf{A} : (D, N) \leftarrow \text{Parameter}$

▷ Represents structured $N \times N$ matrix

2: $\mathbf{B} : (B, L, N) \leftarrow s_B(x)$

3: $\mathbf{C} : (B, L, N) \leftarrow s_C(x)$

4: $\Delta : (B, L, D) \leftarrow \tau_\Delta(\text{Parameter} + s_\Delta(x))$

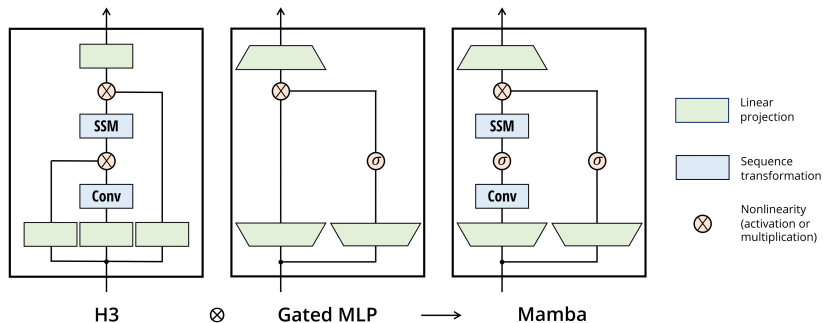
5: $\overline{\mathbf{A}}, \overline{\mathbf{B}} : (B, L, D, N) \leftarrow \text{discretize}(\Delta, \mathbf{A}, \mathbf{B})$

6: $y \leftarrow \text{SSM}(\overline{\mathbf{A}}, \overline{\mathbf{B}}, \mathbf{C})(x)$

▷ Time-varying: recurrence (scan) only

7: **return** y

The Mamba Architecture



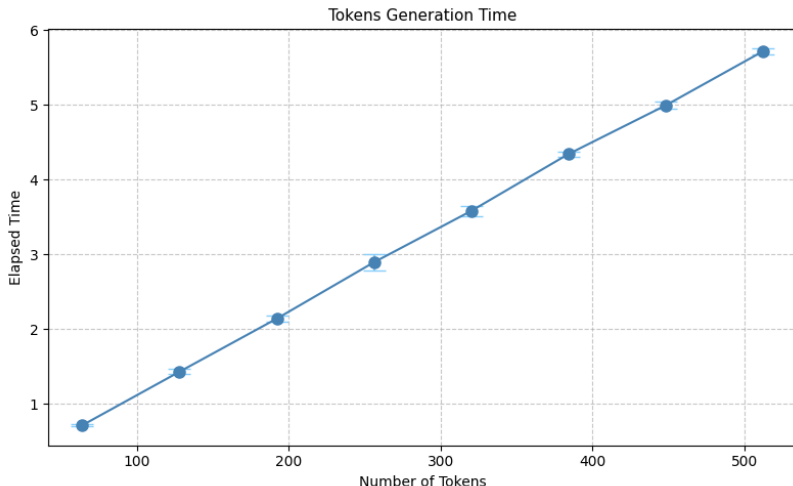
Experiments

We trained a model on the *Divina Commedia* (Alighieri, 1321) with the following Mamba parameters:

- Embedding dim. = 512;
- Expansion factor = 2.5;
- Nr. of channels of the latent state = 16;
- Convolution kernel size = 6;
- Depth = 8.

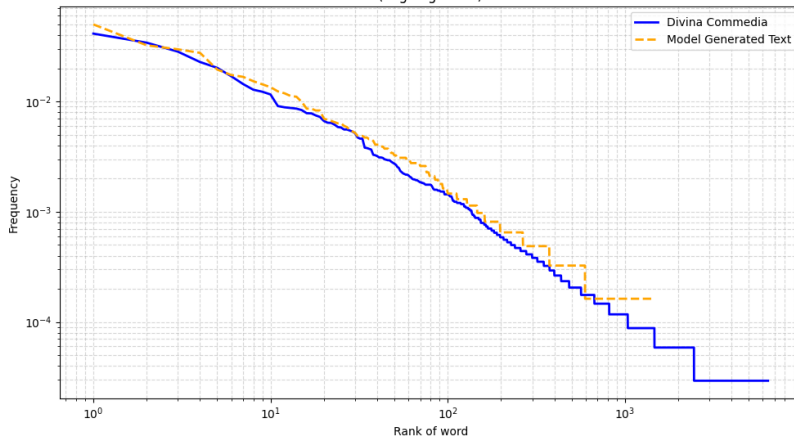
As optimization algorithm we used Adam with 0.01 weight decay, learning rate $1e-3$ with some decay schedule on plateaus, 1024 tokens length and non-constant batch size (4 to 12 depending on the power of the machine).
> 300 iterations were completed.

Despite their worse performance compared to transformers, state space models can generate new text in linear (in L) both time and memory. Transformers require KV -cache (quadratic in L) to generate linearly in L .



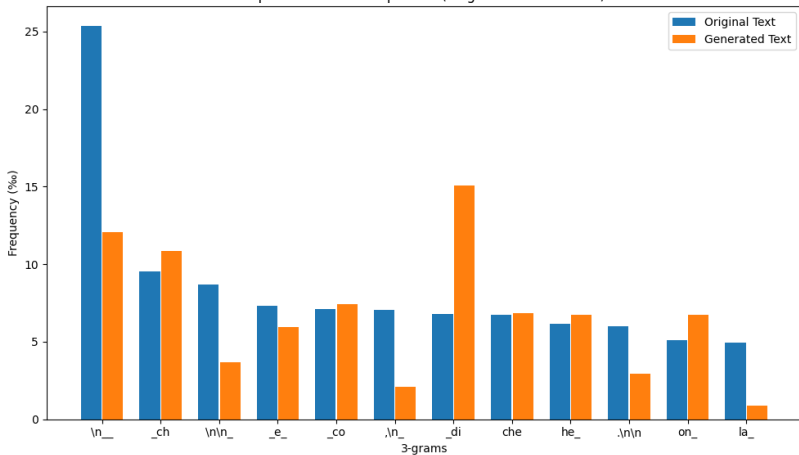
Zipf's Law

Word Frequency Distribution
(Log-Log Scale)



n -Grams Distribution Comparison

Top 12 3-Grams Comparison (Original vs. Generated)



Perplexity on Different Texts

Evaluation of how much the model is surprised by diverse texts.

	Loss	Perplexity
Petrarca	6.8289	924.18
Wikipedia	32.4686	2.5e13
Generated	0.9660	2.63
Commedia	4.8246	124.54

The loss is just the cross-entropy loss, *perplexity* is instead a measure of confidence defined as

$$\text{Perplexity} = \exp \left(\frac{1}{L} \sum_{i=1}^L \log p(w_i \mid w_{i-1}, w_{i-2}, \dots) \right).$$

It is reasonable to think that many of the characters – like digits – of the Wikipedia article about Dante Alighieri do not appear anywhere in the other text corpora, hence the huge numbers.

Generation Heuristics

Temp. T (probs. = $\text{Softmax}(\text{logits}/T)$), top- K (K most likely tokens) and top- P (tokens whose probs. sum up to P) lead to more rigid or fancier outputs.

Cold style

$T=0.3, K=10, P=0.9$

Lasciate ogni speranza
non torre tosto quei poz-
zo che m'ha / crido questo
imorgo giuso disse: "Tu sì
gran di suol che son veg-
gio fossi era più che tu
sì gran di suol d'ogne ga-
rogne giardie". // "An-
crimo, e risi termi, che
'l morto combia tegger-
die on fessi è mi solui che
s'appresso de l'argolia", /

Normal style

$T=1, K=15, P=0.99$

Lasciate ogni speran di sé
già m'accorse e fu, perché
non torrgo dridole ond' el
fossi e quelli a me: "Or
tunge giuggia per troda,
"qua s'a 'nfurata; / e io
'n lui pagna, prima me:
"Quando più di volge di-
sto; / e vidi se 'l pibica così
fattina mi più d'è vel som-
mo comio a tutto li face
monci", /

Warm style

$T=3, K=n/a, P=1$

Lasciate ogni speran se lo-
co for de questi veggio. //
Ma dir discogli. // Mento
vistagni; / ne Al non quivi
rispoi farmi, di ch'è la ol-
tuso me qual ch'a troverza
m'erbol' io vien cesan ca-
da li ha alcun piedue omai
rispuose 'l pociò che pie-
to ver divilliugne 'l testale
ch'usire fondo soglio". //

References



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